

Computer Graphics and Animation

BCA — Semester V — 2022 — Answer Sheet

Subject Code: CACS305

Note: This answer sheet is currently a template. Detailed answers will be added progressively. Students are encouraged to refer to textbooks and class notes.

Group B

2. Define horizontal and vertical retrace of electron beam. Explain architecture of random scan display system.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

3. How does decision parameter can be used to draw circle? Calculate the points to draw a circle having radius 5 and center has (10, 5) using mid-point circle algorithm.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

4. What do you mean by window to viewport transformation? Explain 2D viewing pipeline.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

5. Generate the points between the end points of a line viz. (2, 2) and (9, 6) by using DDA line drawing algorithm.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

6. Define visible surface detection. Outline the Z-buffer algorithm along with its advantages and disadvantages.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

7. What do you mean by Virtual Reality? Explain the types of Virtual Reality.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

8. Define reflection transformation and derive the 2D reflection matrix along x-axis and y-axis in homogeneous coordinate.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

Group C

Attempt any TWO questions[2x10=20]

9. What is the need of homogeneous coordinate system in geometric transformation system? Find the new co-ordinate of rectangle ABCD whose center is at (4, 2) is reduced to half of its size and center will remain same. The co-ordinate of ABCD are A(0, 0), B(0, 4), C(8, 4) and D(8, 0).

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

10. Explain Sutherland-Hodgeman Polygon Clipping algorithm and trace it to clip a polygon with end points P(10, 30), Q(25, 35), R(35, 50), S(60, 30), T(45, 28), U(40, 15) against a window whose lower left corner is at (15, 25) and upper right corner is at (50, 40).

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

11. What do you understand by object space and image space method? Explain Flood Filled algorithm with its applicability and concept.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

These answer sheets are prepared for educational purposes. Students are encouraged to verify with official textbooks and course materials.